

A Monero data model and API specification

*with implementations in Python, Java, TypeScript, and C++
using monero v0.18.5.1 'Flourine Fermi'*

Last updated July 28, 2026



Python implementation: <https://github.com/everoddandeven/monero-python>

Java implementation: <https://github.com/woodser/monero-java>

TypeScript implementation: <https://github.com/woodser/monero-ts>

C++ implementation: <https://github.com/woodser/monero-cpp>

Document source: <https://github.com/everoddandeven/monero-python/blob/master/docs/monero-spec.drawio>

Document created and revisable using: <https://www.draw.io/>

Monero Daemon Interface 1/2

set_proxy_uri(proxy_uri: str) -> None

is_connected() -> bool

get_version() -> MoneroVersion

is_trusted() -> bool

get_height() -> int

get_block_hash(height: int) -> str

get_block_template(wallet_address: str, reserve_size: Optional[int]=None) -> MoneroBlockTemplate

get_last_block_header() -> MoneroBlockHeader

get_block_header_by_hash(blockHash: str) -> MoneroBlockHeader

get_block_header_by_height(height: int) -> MoneroBlockHeader

get_block_headers_by_range(start_height: int, end_height: int) -> list[MoneroBlockHeader]

get_block_by_hash(block_hash: str) -> MoneroBlock

get_blocks_by_hash(block_hashes: list[str], start_height: int, prune: bool=False) -> list[MoneroBlock]

get_block_by_height(height: int) -> MoneroBlock

get_blocks_by_height(heights: list[int]) -> list[MoneroBlock]

get_blocks_by_range(start_height: int, end_height: int) -> list[MoneroBlock]

get_blocks_by_range_chunked(start_height: Optional[int]=None, end_height: Optional[int]=None, max_chunk_size: Optional[int]=None) -> list[MoneroBlock]

get_block_hashes(block_hashes: list[str], start_height: int) -> list[str]

get_tx(tx_hash: str, prune: bool=False) -> MoneroTx

get_txs(tx_hashes: list[str], prune: bool=False) -> list[MoneroTx]

get_tx_hex(tx_hash: str, prune: bool=False) -> str

get_tx_hexes(tx_hashes: list[str], prune: bool=False) -> list[str]

get_miner_tx_sum(height: int, num_blocks: int) -> MoneroMinerTxSum

get_fee_estimate(grace_blocks: int) -> MoneroFeeEstimate

submit_tx_hex(tx_hex: str, do_not_relay: bool=False) -> MoneroSubmitTxResult

relay_tx_by_hash(tx_hash: str) -> None

relay_txs_by_hash(tx_hashes: list[str]) -> None

get_tx_pool() -> list[MoneroTx]

get_tx_pool_hashes() -> list[str]

get_tx_pool_backlog() -> list[MoneroTxBacklogEntry]

get_tx_pool_stats() -> MoneroTxPoolStats

flush_tx_pool(txHashes: Optional[list[str]]=None) -> None

get_key_image_spent_status(key_image: str) -> MoneroKeyImageSpentStatus

get_key_image_spent_statuses(key_images: list[str]) -> list[MoneroKeyImageSpentStatus]

get_outputs(outputs: list[MoneroOutput]) -> list[MoneroOutput]

get_info() -> MoneroDaemonInfo

get_sync_info() -> MoneroDaemonSyncInfo

get_hard_fork_info() -> MoneroHardForkInfo

Monero Daemon Interface 2/2

get_alt_chains() -> list[MoneroAltChain]
get_alt_block_hashes(): list[str]
get_download_limit(): int
set_download_limit(limit: int): int
reset_download_limit() -> int
get_upload_limit() -> int
set_upload_limit(limit: int) -> int
reset_upload_limit() -> int
get_peers(): list[MoneroPeer]
get_known_peers() -> list[MoneroPeer]
set_outgoing_peer_limit(limit: int) -> None
set_incoming_peer_limit(limit: int) -> None
get_peer_bans(): list[MoneroBan]
set_peer_ban(ban: MoneroBan) -> None
set_peer_bans(bans: list[MoneroBan]) -> None
get_output_distribution(amounts: list[int], is_cumulative: bool, start_height: int=0, end_height: Optional[int]=None) -> list[MoneroOutputDistributionEntry]
get_output_histogram(amounts: list[int], min_count: int, max_count: int, is_unlocked: bool, recent_cutoff: int) -> list[MoneroOutputHistogramEntry]
start_mining(address: str, num_threads: int, is_background: bool=False, ignore_battery: bool=False) -> None
stop_mining() -> None
get_mining_status() -> MoneroMiningStatus
submit_block(block_blob: str) -> None
submit_blocks(block_blobs: list[str]) -> None
check_for_update() -> MoneroDaemonUpdateCheckResult
download_update(path: str) -> MoneroDaemonUpdateDownloadResult
add_listener(listener: MoneroDaemonListener) -> None
remove_listener(listener: MoneroDaemonListener) -> None
wait_for_next_block_header() -> MoneroBlockHeader
stop() -> None
prune_blockchain(check: bool) -> MoneroPruneResult

Monero Daemon Types 1/3

MoneroBlockHeader
hash: Optional[str]
height: Optional[int]
timestamp: Optional[int]
size: Optional[int]
weight: Optional[int]
long_term_weight: Optional[int]
depth: Optional[int]
difficulty: Optional[int]
cumulative_difficulty: Optional[int]
major_version: Optional[int]
minor_version: Optional[int]
nonce: Optional[int]
miner_tx_hash: Optional[str]
num_txs: Optional[int]
orphan_status: Optional[bool]
prev_hash: Optional[str]
reward: Optional[int]
pow_hash: Optional[str]



Extends

MoneroBlock
hex: Optional[str]
miner_tx: Optional[MoneroTx]
txs: list[MoneroTx]
tx_hashes: list[str]

MoneroOutput
tx: Optional[MoneroTx]
key_image: Optional[MoneroKeyImage]
amount: Optional[int]
index: Optional[int]
ring_output_indices: list[int]
stealth_public_key: Optional[str]

MoneroDaemonInfo
version: Optional[str]
num_alt_blocks: Optional[int]
block_size_limit: Optional[int]
block_size_median: Optional[int]
block_weight_limit: Optional[int]
block_weight_median: Optional[int]
bootstrap_daemon_address: Optional[str]
difficulty: Optional[int]
cumulative_difficulty: Optional[int]
free_space: Optional[int]
num_offline_peers: Optional[int]
num_online_peers: Optional[int]
height: Optional[int]
height_without_bootstrap: Optional[int]
network_type: Optional[MoneroNetworkType]
is_offline: Optional[bool]
num_incoming_connections: Optional[int]
num_outgoing_connections: Optional[int]
num_rpc_connections: Optional[int]
start_timestamp: Optional[int]
adjusted_timestamp: Optional[int]
target: Optional[int]
target_height: Optional[int]
top_block_hash: Optional[str]
num_txs: Optional[int]
num_txs_pool: Optional[int]
was_bootstrap_ever_used: Optional[bool]
database_size: Optional[int]
update_available: Optional[bool]
credits: Optional[int]
is_busy_syncing: Optional[bool]
is_synchronized: Optional[bool]
is_restricted: Optional[bool]

MoneroTx
block: Optional[MoneroBlock]
height: Optional[int]
hash: Optional[str]
version: Optional[int]
is_coinbase: Optional[bool]
payment_id: Optional[str]
fee: Optional[int]
ring_size: Optional[int]
do_not_relay: Optional[bool]
is_relayed: Optional[bool]
is_confirmed: Optional[bool]
in_tx_pool: Optional[bool]
num_confirmations: Optional[int]
unlock_time: Optional[int]
last_relayed_timestamp: Optional[int]
received_timestamp: Optional[int]
is_double_spend: Optional[bool]
key: Optional[str]
full_hex: Optional[str]
pruned_hex: Optional[str]
prunable_hex: Optional[str]
prunable_hash: Optional[str]
size: Optional[int]
weight: Optional[int]
inputs: list[MoneroOutput]
outputs: list[MoneroOutput]
output_indices: list[int]
metadata: Optional[str]
extra: list[int]
rct_signatures: list[str]
rct_sig_prunable: Optional[str]
is_kept_by_block: Optional[bool]
is_failed: Optional[bool]
last_failed_height: Optional[int]
last_failed_hash: Optional[str]
max_used_block_height: Optional[int]

Monero Daemon Types 2/3

max_used_block_hash: Optional[str]

signatures: list[str]

MoneroDaemonSyncInfo

height: Optional[int]
peers: list[MoneroPeer]
spans: list[MoneroConnectionSpan]
target_height: Optional[int]
next_needed_pruning_seed: Optional[int]
overview: Optional[str]
top_block_hash: Optional[str]
credits: Optional[int]

MoneroBan

host: Optional[str]
ip: Optional[str]
is_banned: Optional[bool]
seconds: Optional[int]

MoneroBlockTemplate

block_template_blob: Optional[str]
block_hashing_blob: Optional[str]
difficulty: Optional[int]
expected_reward: Optional[int]
height: Optional[int]
prev_hash: Optional[str]
reserved_offset: Optional[int]
seed_height: Optional[int]
seed_hash: Optional[str]
next_seed_hash: Optional[str]

MoneroConnectionSpan

connection_id: Optional[str]
num_blocks: Optional[int]
remote_address: Optional[str]
rate: Optional[int]
speed: Optional[int]
size: Optional[int]
start_block_height: Optional[int]

MoneroDaemonListener

on_block_header(header: MoneroBlockHeader) -> None

<enumeration> MoneroNetworkType

MAINNET: 0
TESTNET: 1
STAGENET: 2

MoneroKeyImage

hex: Optional[str]
signature: Optional[str]

<enumeration> MoneroKeyImageSpentStatus

NOT_SPENT: 0
CONFIRMED: 1
TX_POOL: 2

MoneroSubmitTxResult

is_good: Optional[bool]
is_relayed: Optional[bool]
is_double_spend_seen: Optional[bool]
is_fee_too_low: Optional[bool]
is_mixin_too_low: Optional[bool]
has_invalid_input: Optional[bool]
has_invalid_output: Optional[bool]
has_too_few_outputs: Optional[bool]
is_rct: Optional[bool]
is_overspend: Optional[bool]
is_too_big: Optional[bool]
sanity_check_failed: Optional[bool]
reason: Optional[str]
credits: Optional[int]
top_block_hash: Optional[str]
is_tx_result_too_big: Optional[bool]
nonzero_unlock_time: Optional[bool]

MoneroPeer

id: Optional[str]
address: Optional[str]
host: Optional[str]
port: Optional[int]
is_online: Optional[bool]
last_seen_timestamp: Optional[int]
pruning_seed: Optional[int]
rpc_port: Optional[int]
rpc_credits_per_hash: Optional[int]
type: Optional[MoneroConnectionType]
num_support_flags: Optional[int]
state: Optional[str]
send_idle_time: Optional[int]
receive_idle_time: Optional[int]
num_sends: Optional[int]
num_receives: Optional[int]
is_local_host: Optional[bool]
is_local_ip: Optional[bool]
live_time: Optional[int]
is_incoming: Optional[bool]
height: Optional[int]
current_upload: Optional[int]
current_download: Optional[int]
avg_upload: Optional[int]
avg_download: Optional[int]

Monero Daemon Types 3/3

MoneroHardForkInfo
earliest_height: Optional[int]
is_enabled: Optional[bool]
state: Optional[int]
threshold: Optional[int]
version: Optional[str]
num_votes: Optional[int]
window: Optional[int]
voting: Optional[int]
credits: Optional[int]
top_block_hash: Optional[str]

MoneroVersion
number: Optional[int]
is_release: Optional[bool]

<enumeration> MoneroConnectionType
INVALID: 0
IPV4: 1
IPV6: 2
TOR: 3
I2P: 4

MoneroFeeEstimate
fee: Optional[int]
fees: list[int]
quantization_mask: Optional[int]

MoneroDaemonUpdateCheckResult
is_update_available: Optional[bool]
version: Optional[str]
hash: Optional[str]
auto_uri: Optional[str]
user_uri: Optional[str]



MoneroDaemonUpdateDownloadResult
download_path: Optional[str]

MoneroMinerTxSum
emission_sum: Optional[int]
fee_sum: Optional[int]

MoneroMiningStatus
is_active: Optional[bool]
is_background: Optional[bool]
address: Optional[str]
speed: Optional[int]
num_threads: Optional[int]

MoneroPruneResult
is_pruned: Optional[bool]
pruning_seed: Optional[int]

MoneroOutputHistogramEntry
amount: Optional[int]
num_instances: Optional[int]
num_unlocked_instances: Optional[int]
num_recent_instances: Optional[int]

MoneroTxPoolStats
num_txs: Optional[int]
num_not_relayed: Optional[int]
num_failing: Optional[int]
num_double_spends: Optional[int]
num_10m: Optional[int]
fee_total: Optional[int]
bytes_max: Optional[int]
bytes_med: Optional[int]
bytes_min: Optional[int]
bytes_total: Optional[int]
histo_98pc: Optional[int]
oldest_timestamp: Optional[int]

MoneroAltChain
block_hashes: list[str]
difficulty: Optional[int]
height: Optional[int]
length: Optional[int]
main_chain_parent_block_hash: Optional[str]

Monero Wallet Interface 1/3

add_listener(listener: MoneroWalletListener) -> None

remove_listener(listener: MoneroWalletListener) -> None

get_listeners() -> list[MoneroWalletListener]

is_view_only() -> bool

set_daemon_connection(connection: MoneroDaemonConnection) -> None

get_daemon_connection() -> MoneroDaemonConnection

is_connected_to_daemon() -> bool

get_version() -> MoneroVersion

get_network_type() -> MoneroNetworkType

get_path() -> str

get_seed() -> str

get_seed_language() -> str

get_seed_languages() -> list[str]

get_private_spend_key() -> str

get_private_view_key() -> str

get_public_spend_key() -> str

get_public_view_key() -> str

get_primary_address() -> str

get_address(account_idx: int, subaddress_idx: int) -> str

get_address_index(address: str) -> MoneroSubaddress

get_daemon_connection() -> MoneroRpcConnection

get_daemon_height() -> int

get_daemon_max_peer_height() -> int

get_height() -> int

get_height_by_date(year: int, month: int, day: int) -> int

sync(startHeight: Optional[int]=None, listener: Optional[MoneroSyncListener]=None) -> MoneroSyncResult

start_syncing() -> None

stop_syncing() -> None

scan_txs(tx_hashes: list[str]) -> None

rescan_spent() -> None

rescan_blockchain() -> None

get_balance(account_idx: Optional[int]=None, subaddress_idx: Optional[int]=None) -> int

get_unlocked_balance(account_idx: Optional[int]=None, subaddress_idx: Optional[int]=None) -> int

get_accounts(include_subaddresses: bool=False, tag: Optional[str]=None) -> list[MoneroAccount]

get_account(account_idx: int, include_subaddresses: bool=False) -> MoneroAccount

create_account(label: Optional[str]=None) -> None

get_subaddresses(account_idx: int, subaddress_indices: list[int]) -> None

get_subaddress(account_idx: int, subaddress_idx: int) -> MoneroSubaddress

Monero Wallet Interface 2/3

create_subaddress(account_idx: int, label: Optional[str]=None) -> MoneroSubaddress

set_account_label(account_idx: int, label: str) -> None

set_subaddress_label(account_idx: int, subaddress_idx: int, label: str) -> None

get_tx(tx_id: str) -> MoneroTxWallet

get_txs(query Optional[MoneroTxQuery]=None) -> list[MoneroTxWallet]

get_transfers(account_idx: Optional[int]=None, subaddressIdx: Optional[int]=None) -> list[MoneroTransfer]

get_outputs(query: Optional[MoneroOutputQuery]=None) -> list[MoneroOutputWallet]

get_outgoing_transfers(query: Optional[MoneroTransferQuery]=None) -> list[MoneroOutgoingTransfer]

get_incoming_transfers(query: Optional[MoneroTransferQuery]=None) -> list[MoneroIncomingTransfer]

get_transfers(query: Optional[MoneroTransferQuery]=None) -> None

export_outputs(all: bool=False) -> str

import_outputs(outputs_hex: str) -> int

export_key_images(all: bool=False) -> list[MoneroKeyImage]

import_key_images(keyImages: list[MoneroKeyImage]) -> MoneroKeyImageImportResult

freeze_output(key_image: str) -> None

thaw_output(key_image: str) -> None

is_output_frozen(key_image: str) -> bool

get_new_key_images_from_last_import() -> list[MoneroKeyImage]

get_default_fee_priority() -> MoneroTxPriority

create_tx(config: MoneroTxConfig) -> MoneroTxWallet

create_txs(config: MoneroTxConfig) -> list[MoneroTxWallet]

sweep_output(config: MoneroTxConfig) -> MoneroTxWallet

sweep_unlocked(config: MoneroTxConfig) -> list[MoneroTxWallet]

sweep_dust(relay: bool=False) -> list[MoneroTxWallet]

relay_tx(tx_metadata: str) -> str

relay_tx(tx: MoneroTxWallet) -> str

relay_txs(tx_metadata: list[str]) -> list[str]

relay_txs(txs: list[MoneroTxWallet]) -> list[str]

describe_tx_set(tx_set: MoneroTxSet) -> MoneroTxSet

sign_txs(unsigned_tx_hex: str) -> MoneroTxSet

submit_txs(signed_tx_hex: str) -> list[str]

sign_message(message: str, type: MoneroMessageSignatureType, account_idx: int, subaddress_idx: int) -> str

verify_message(message: str, address: str, signature: str) -> MoneroMessageSignatureResult

get_tx_key(tx_id: str) -> str

check_tx_key(tx_id: str, tx_key: str, address: str) -> MoneroCheckTx

get_tx_proof(tx_id: str, address: str, message: Optional[str]=None) -> str

check_tx_proof(tx_id: str, address: str, message: str, signature: str) -> MoneroCheckTx

get_spend_proof(tx_id: str, message: Optional[str]=None) -> str

check_spend_proof(tx_id: str, message: str, signature: str) -> bool

Monero Wallet Interface 3/3

```
get_reserve_proof_wallet(message: str) -> str

get_reserve_proof_account(account_idx: int, amount: int, message: str) -> str

check_reserve_proof(address: str, message: str, signature: str) -> MoneroCheckReserve

set_tx_notes(tx_hashes: list[str], notes: list[str]) -> None

set_tx_note(tx_hash: str, note: str) -> None

get_tx_notes(tx_hashes: str): list[str]

get_tx_note(tx_hash: str) -> str

get_address_book_entries(entryIndices: list[int]=[]) -> list[MoneroAddressBookEntry]

add_address_book_entry(address: str, description: str) -> int

edit_address_book_entry(entry_idx: int, set_address: bool, address: str, set_description: bool, description: str) -> None

delete_address_book_entry(entry_idx: int) -> None

tag_accounts(tag: str, account_indices: list[int]) -> None

untag_accounts(account_indices: list[int]) -> None

get_account_tags() -> list[MoneroAccountTag]

set_account_tag_label(tag: str, label: str) -> None

get_payment_uri(request: MoneroTxConfig) -> str

parse_payment_uri(uri: str) -> MoneroSendRequest

get_attribute(key: str) -> str

set_attribute(key: str, val: str) -> None

start_mining(uint num_threads: int, background_mining: bool=False, ignore_battery: bool=True) -> None

stop_mining() -> None

is_multisig_import_needed() -> bool

is_multisig() -> bool

get_multisig_info() -> MoneroMultisigInfo

prepare_multisig() -> str

make_multisig(multisig_hexes: list[str], threshold: int, password: str) -> str

exchange_multisig_keys(multisig_hexes: list[str], password: str) -> MoneroMultisigInitResult

export_multisig_hex() -> str

import_multisig_hex(multisig_hexes: list[str]) -> int

sign_multisig_tx_hex(multisig_tx_hex: str) -> MoneroMultisigSignResult

submit_multisig(signed_multisig_hex: str) -> list[str]

move_to(path: str) -> None

change_password(old_password: str, new_password: str) -> None

save() -> None

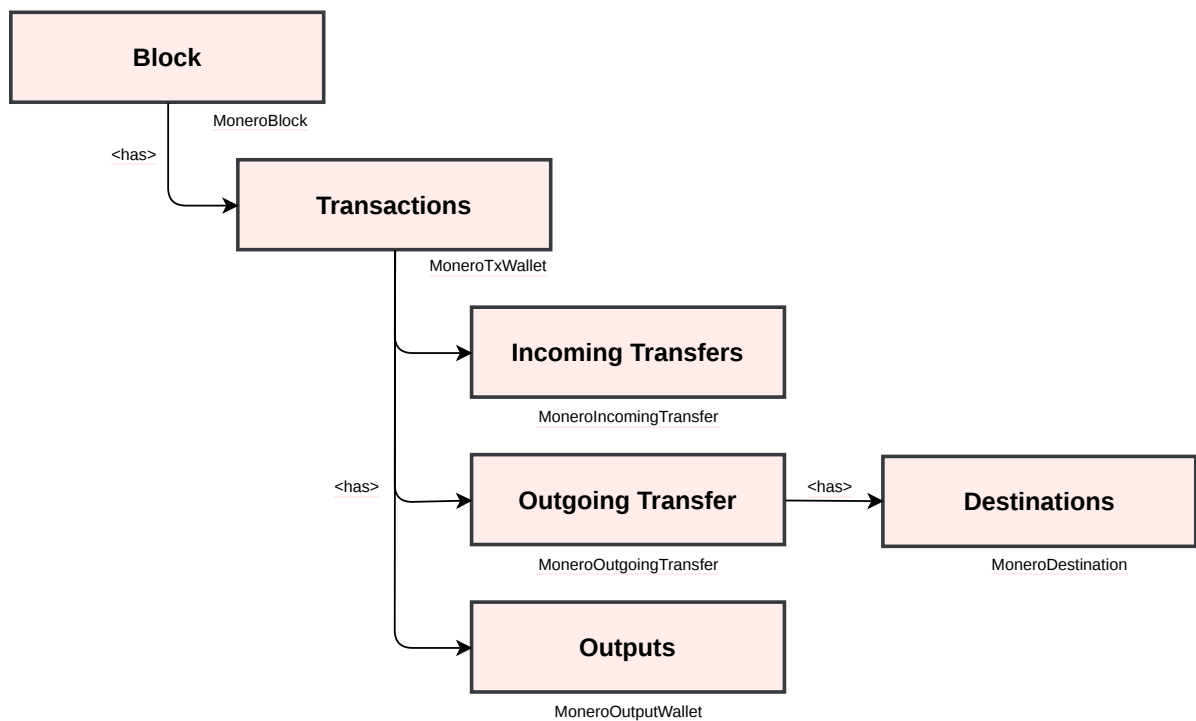
close(save: bool=false) -> None

is_closed() -> bool

get_integrated_address(standardAddress: Optional[str]=None, payment_id: Optional[str]=None) -> MoneroIntegratedAddress (deprecated)

decode_integrated_address(integrated_address: str) -> MoneroIntegratedAddress (deprecated)
```

Monero Wallet Types 1/4 - Transactions, Transfers, and Outputs



Transactions are part of a block and have incoming transfers and/or an outgoing transfer.

The outgoing transfer may have destination amounts and addresses which are saved locally with the wallet and are not recoverable from the blockchain.

Transactions may also have outputs received by the wallet when requested.

Monero Wallet Types 2/4 - Send Request and Queries for Transactions, Transfers, and Outputs

MoneroTxConfig
address: Optional[str]
amount: Optional[int]
destinations: list[MoneroDestination]
subtract_fee_from: list[int]
payment_id: Optional[str]
priority: Optional[MoneroTxPriority]
fee: Optional[int]
account_index: Optional[int]
subaddress_index: Optional[int]
subaddress_indices: list[int]
can_split: Optional[bool]
relay: Optional[bool]
note: Optional[str]
recipient_name: Optional[str]
below_amount: Optional[int]
sweep_each_subaddress: Optional[bool]
key_image: Optional[str]

Configures creating transactions to send funds.

MoneroTxQuery extends MoneroTxWallet
is_outgoing: Optional[bool]
is_incoming: Optional[bool]
hashes: list[str]
has_payment_id: Optional[bool]
payment_ids: list[str]
height: Optional[int]
min_height: Optional[int]
max_height: Optional[int]
include_outputs: Optional[bool]
transfer_query: Optional[MoneroTransferQuery]
input_query: Optional[MoneroOutputQuery]
output_query: Optional[MoneroOutputQuery]

Configures a query to get wallet transactions.

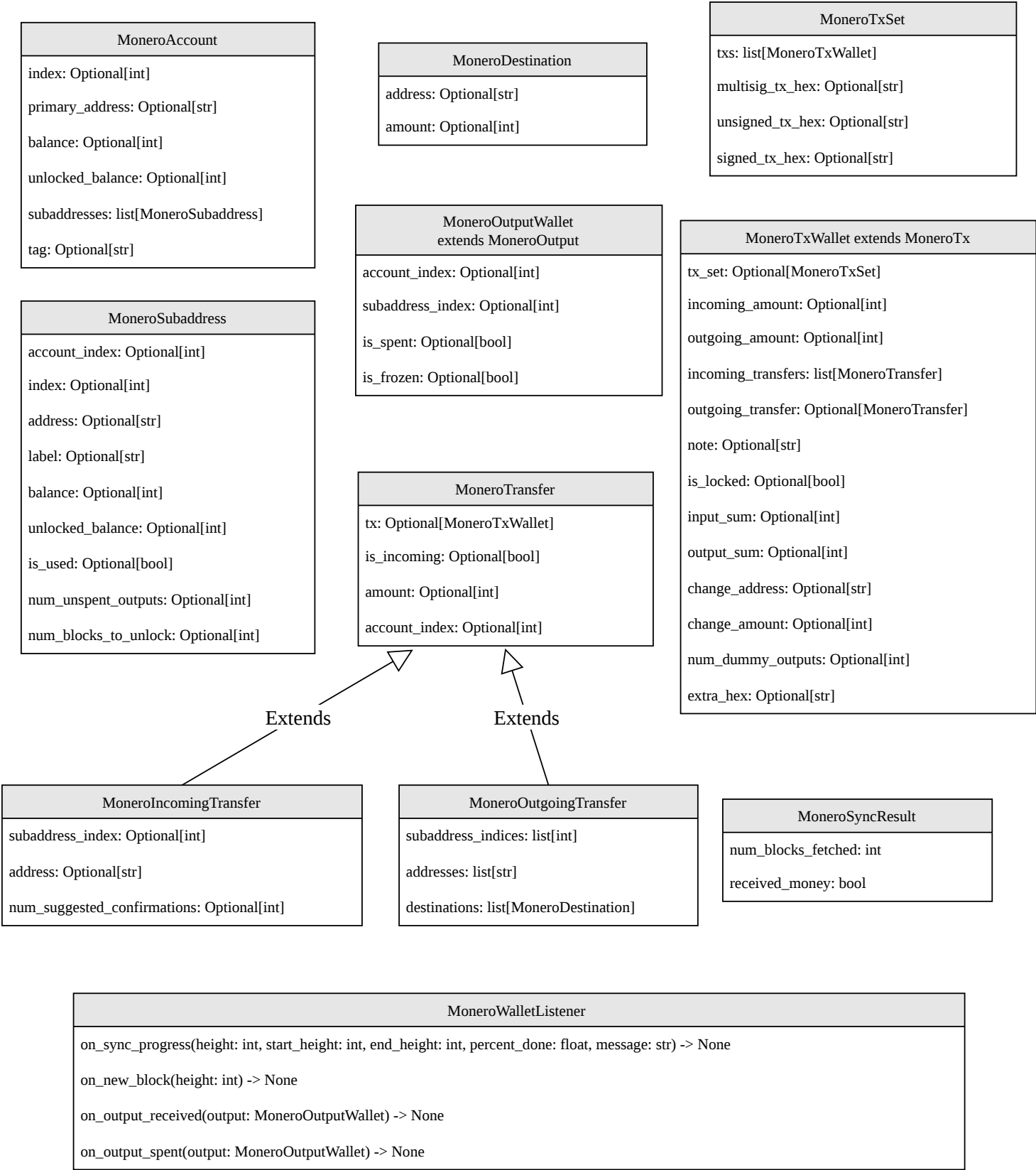
MoneroOutputQuery extends MoneroOutputWallet
subaddress_indices: list[int]
tx_query: Optional[MoneroTxQuery]
min_amount: Optional[int]
max_amount: Optional[int]

Configures a query to get wallet outputs.

MoneroTransferQuery extends MoneroTransfer
is_incoming: Optional[bool]
address: Optional[str]
addresses: list[str]
subaddress_index: Optional[int]
subaddress_indices: list[int]
destinations: list[MoneroDestination]
has_destinations: Optional[bool]
tx_query: Optional[MoneroTxQuery]

Configures a query to get wallet transfers.

Monero Wallet Types 3/4



Monero Wallet Types 4/4

MoneroAccountTag
tag: Optional[str]
label: Optional[str]
account_indices: list[int]

MoneroMultisigInfo
is_multisig: bool
is_ready: bool
threshold: int
num_participants: int

MoneroKeyImageImportResult
height: Optional[int]
spent_amount: Optional[int]
unspent_amount: Optional[int]

MoneroAddressBookEntry
index: Optional[int]
address: Optional[str]
description: Optional[str]
payment_id: Optional[str]

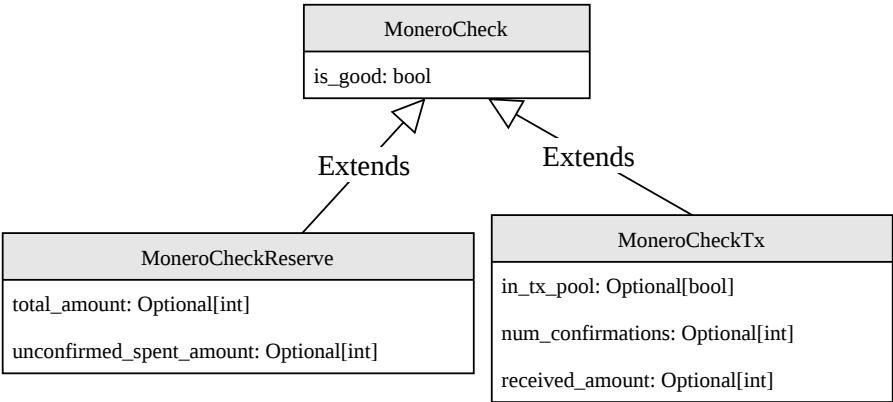
MoneroMultisigInitResult
address: Optional[str]
multisig_hex: Optional[str]

MoneroIntegratedAddress (deprecated)
standard_address: str
payment_id: str
integrated_address: str

<<enumeration>> MoneroSendPriority
DEFAULT: 0
UNIMPORTANT: 1
NORMAL: 2
ELEVATED: 3

MoneroMultisigSignResult
signed_multisig_tx_hex: Optional[str]
tx_hashes: list[str]

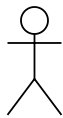
<<enumeration>> MoneroMessageSignatureType
SIGN_WITH_SPEND_KEY: 0
SIGN_WITH_VIEW_KEY: 1



MoneroMessageSignatureResult
is_good: bool
is_old: bool
signature_type: MoneroMessageSignatureType
version: int



84rXAU3QhsfbQ2vbxyoEptbRLZ2gtyL8k2qCeJwM5AU4FcxVDEbLRuxMZD5pXD1TR295Nodn7y9mkjoMemHrnMqa9ZnJTza



everoddandeven
donation_address: const string
matrix: "everoddandeven"
reddit: "everoddandeven"
time_committed_to_xmr: ulong
competing_interests: ["GIS"]

Motivation tends to increase with support to donation address.